Pankaj Rana

Address Rutesheim, Germany
Phone Number +49 15783445366
E-Mail pankajrana1418@gmail.com
Nationality Indian (Student Visa)
D.O.B 14 September, 1996

LinkedIn

LeetCode

Professional Experience

May 2025 – Feb. 2026

Master's Thesis Student at Zentrum für Sonnenenergie- und Wasserstoff-Forschung Baden-Württemberg (ZSW)

Project Real-time Bat detection on wind turbines | 3D Object detection with AI

- Calibrated and integrated LiDAR and near-infrared cameras for recording and analyzing nocturnal flight movements.
- Developed a ROS2-based pipeline for 3D point cloud preprocessing, including adaptive noise filtering and background subtraction (NumPy, Open3D).
- Implemented and evaluated clustering-based 3D object detection methods and multi-object tracking.
- Built and analyzed datasets from real-world measurements and simulation-based scenarios, applying deep learning for spatio-temporal motion analysis.

Aug. 2024 – Feb. 2026

Research Assistant at the Chair of Dynamics and Control (SRS), University of Duisburg-Essen

Computer Science, DSA, and Programming

- Conceptualized and developed programming exercises in Data Structures and Algorithms (DSA).
- Implemented solutions in Python, emphasizing algorithmic efficiency, scalability, and clean coding practices.
- Designed and executed pytest test frameworks to validate algorithmic problem solutions, ensuring correctness across diverse edge cases.

Nov. 2024 – April 2025

Software Development Intern at Porsche Engineering Services

Project ADAS Training Simulator | Software Development and Autonomous Driving

- Developed a Windows application in C++ to control the steering wheel, seat actuator, and other hardware components of the ADAS training simulator.
- Implementation of control loops for precise control of movements based on real-time sensor data in the CARLA simulation environment.
- Worked in an agile development environment following Scrum and Kanban methodologies to ensure iterative delivery and system reliability.

June 2024 - Sept. 2024

Research Assistant at DST - Development Center for Ship Technology and Transport Systems e.V.

Project Cobotank | Robotics

- Implemented coordinate transformations in ROS2 using C++ and Python to enable accurate multi-sensor alignment.
- Created and optimized 3D environment representations, integrating camera model into the robot model for perception tasks.
- Contributed to system debugging and testing, ensuring reliability of robotics software modules.

Education

April 2022 - Present

Master's Degree (M.Sc.) University of Duisburg-Essen, Germany

Mechanical Engineering (Specialization in Mechatronics)

- Machine Learning & Artificial Intelligence
- Programming & Data Analysis
- Robotics & Control Systems
- Current grade: 2.1

Projects

April 2024 – Oct. 2024

Traffic and Battery status analysis using Machine Learning

University of Duisburg-Essen

- **Traffic data classification:** Implemented and compared machine learning classifiers (e.g., Decision Trees, Random Forests) for speed analysis and flow category classification using Python with scikit-learn and Pandas.
- **Battery health prediction:** Predicted battery State of Health (SoH) and Remaining Useful Life (RUL) using the NASA Li-ion dataset, leveraging Python, NumPy, and scikit-learn for data processing, feature engineering, and model development.
- Gained experience in data preprocessing, supervised learning, and performance benchmarking.

Oct. 2023 - Jan 2024

Robotics Application - Board Plotter

University of Duisburg-Essen

- Programmed the ESP32 microcontroller in C++ to control a board-mounted cable-driven plotter.
- Development and implementation of precise motion logic and algorithms.
- Configuration, calibration and ensuring the accuracy of the motors.
- Integration of safety mechanisms and error handling.
- Carrying out tests and optimizations to improve performance.

Aug. 2016 - July 2020

Bachelor of Engineering (B.E.) Panjab University, India

Mechanical Engineering

- Control Systems & Programming Fundamentals
- Final grade: 2.25

Skills

Language Skills — **German** (B2), **English** (C2)

Programming Languages
- Python, C++

Machine Learning & Data
- PyTorch, TensorFlow, scikit-learn, Open3D, OpenCV, Point Cloud Library (PCL)

Simulation & Tools
- Gazebo, CARLA (Basics), Git, pytest

Robotics & Systems
- ROS2, Linux, Bash, Docker (Basics)

Methoden
- V-Modell, Agile (Scrum, Kanban)